Name - Adrija Ghosh

Profit & Loss P~~&~~Q

Q1)

CP	MP	SP
100	120	$120 \times \frac{9}{10} = 108$

$8 \rightarrow 4.8 \overset{+2}{\cancel{4.8}} 0.6$

$100 \rightarrow \frac{\cancel{4.8}}{\cancel{8}_{2}} \times 100 = ₹60$ (Ans)

Q2)

MP	CP	SP
100	90	$^{4}\cancel{18} \cancel{90} \times \frac{6}{\cancel{5}}$

$= 108$

Ans: 8%.

Q3)

₹20 ₹30

26

₹ ~~26~~

2~~4~~ : ~~6~~3

2 : 3

$SP = CP \times mf$

$^{13}\cancel{39} = x \times \frac{3}{2}$

$x = 26$

$2 \rightarrow 26$

$3 \rightarrow \frac{26^{13}}{\cancel{2}} \times 3 = 39$ kg (Ans)

Q4) $21\,SP - 21\,CP = 6\,CP$

$21\,SP = 27\,CP$

$21\,SP = 1620$ (we know)

$CP = \frac{1620}{27}$

60

Ans 60

Q5) $A \xrightarrow{-\frac{1}{5}} R$, $R \xrightarrow{+\frac{1}{5}} A$

$$12000 \times \frac{4}{5} \times \frac{6}{5}$$

$$= 11520 \quad \text{Opt c (Ans)}$$

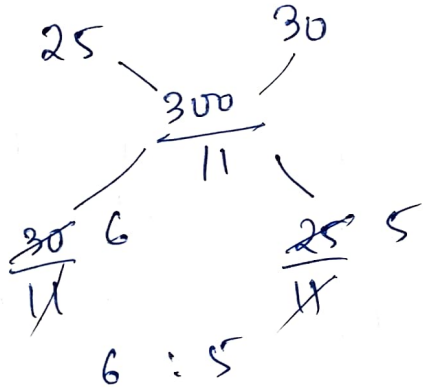
$$\begin{array}{r} 3\ 48 \\ \underline{24} \\ 11\ 52 \end{array} \qquad \begin{array}{r} 3 \\ 16 \\ \underline{16} \\ 35 \end{array}$$

Q6) $-25 \quad -24 + \frac{25 \times 24}{100}$

$$-49 + 6$$

$$= -43\% \text{ discount}$$

Q7)



$SP = CP \times mf$

$$30 = x \times \frac{11}{10}$$

$$\frac{300}{11} - 30 = \frac{300 - 330}{11}$$

$$\frac{300}{11} - 25 = \frac{300 - 275}{11} = 25$$

$5 \to 30$

$6 \to \frac{30}{5} \times 6 = 36 \text{ kg (Ans)}$

Q8) $CP \xrightarrow{+\frac{1}{3}} MP \xrightarrow{-\frac{1}{6}} SP$

$$4500 \times \frac{4}{3} \times \frac{5}{6} = 5000$$

Profit = 500 (Ans)

$$P\% = \frac{500}{4500} \times 100 = \frac{1}{9} = 11.11\%$$

99) Dis = $-20 - 25 + \frac{20 \times 25}{100}$

$= -40\%$

$x \times \frac{60}{100} = 18000$

$x = 30000$ (Ans)

Q10) SP = 110

$$CP = 95 \times \frac{6}{5} = 114$$

4 → 80

100 → $\frac{80}{4} \times 100$

= 2000

Set 02

Q1) 15 CP = 1 12 SP = 1

15 CP = 12 SP

$\frac{CP}{SP} = \frac{12}{15} \; \frac{4}{5}$) 1

Profit = $\frac{1}{4}$ = 25%.

Q2)

CP	MP
100	120

SP = 500 MP = 600

$\frac{1}{6} \times 100$

$= 16\frac{2}{3}\%$ discount

Successive:

$100 \times \frac{5}{6} \times \frac{6}{5}$

Profit = 0%.

3) 1000 gm → 900 gm

CP: ₹1000 → ₹900

Considering 1 gm cost ₹1

CP = ₹900

SP = ₹1000

$\text{Profit} = \frac{100}{900} \times 100$

$= 11.11\%$ gain

4) $CP_1 + CP_2 = 1008$

$SP_1 = SP_2$

$CP_1 \times \frac{4}{5} = CP_2 \times \frac{144}{100}$

$\frac{CP_1}{CP_2} = \frac{144}{100} \times \frac{8}{4}$

$\frac{CP_1}{CP_2} = \frac{36}{20}$

$CP_1 = \frac{36}{56} \times 1008 = 648$ (Ans)

$CP_2 = \frac{20}{56} \times 1008 = 360$ (Ans)

5) $CP_1 + CP_2 = 3000$

$SP_1 = CP_1 \times \frac{122}{100}$ $\quad SP_2 = \frac{92}{100} \times CP_2$

no profit no loss.

$\text{Profit}_1 = \text{Loss}_2$

$CP_1 \times \frac{122}{100} - CP_1 = CP_2 - CP_2 \times \frac{92}{100}$

$CP_1 \times \frac{122}{100} + CP_2 \times \frac{92}{100} = CP_1 + CP_2 = 3000$

$\frac{122\,CP_1}{100} + \frac{92\,CP_2}{100} = 3000$

6) MP SP

300 $300 \times \frac{88}{100} = 264$

CP

$SP = CP \times mf$

$264 = x \times \frac{11}{10}$

$x = 240$

Gain $= \frac{60}{240} \times 100$

$= 25\%$ (Ans)

8-7) $CP_1 + CP_2 = 2940$

$SP_1 = CP_1 \times \frac{120}{100}$ $\quad SP_2 = CP_2 \times \frac{85}{100}$

Overall 5% profit

$SP_1 + SP_2 = \frac{105}{100}(CP_1 + CP_2)$

$\frac{120}{100} CP_1 + \frac{85}{100} CP_2 = \frac{105}{100}(2940)$

$\Rightarrow 120CP_1 + 85CP_2 = 105 \times 2940$